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Course code: CSA1708

Artificial Intelligence CO-1 AT-1

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

Sol:

Problem Understanding

There are two jugs:

- Jug A = 4 gallons
- Jug B = 3 gallons

The goal is to obtain exactly 2 gallons in the 4-gallon jug.

Allowed operations:

- Fill any jug.
- Empty any jug.
- Pour water from one jug to another.

Method Selection

Use State Space Search (Breadth-First Search) because every state can be represented as:

(Amount in 4G Jug, Amount in 3G Jug)

Solution Process

Step 4-Gallon 3-Gallon Action

1	0	0	Start
2	0	3	Fill 3-gallon

Step 4-Gallon 3-Gallon Action

3	3	0	Pour into 4-gallon
4	3	3	Fill 3-gallon again
5	4	2	Pour into 4-gallon until full
6	0	2	Empty 4-gallon
7	2	0	Pour remaining 2 gallons into 4-gallon

Final State

(2, 0)

Exactly 2 gallons are in the 4-gallon jug.

Accuracy of Calculation

Each operation follows the given rules.

Final amount:

4-gallon jug = 2 gallons

Correct.

Interpretation of Results

The Water Jug problem is solved using state-space search, where each state represents water levels in both jugs. The goal state is successfully reached.

2. Find out how Mars Rover, analyze and explore the samples on Mars by answering the following questions.

i. What are the percept's for this agent?

ii. Characterize the operating environment.

iii. What are the actions the agent can take?

iv. How can one evaluate the performance of the agent?

v. What sort of agent architecture do you think is most suitable for this agent?

Why?

Sol:

Problem Understanding

Mars Rover is an autonomous robot that explores Mars, collects samples, analyzes rocks, and sends data back to Earth.

Method Selection

Use the PEAS Model (Performance, Environment, Actuators, Sensors).

Solution Process

i. Precepts

- Camera images
- Rock color
- Soil composition
- Temperature
- Distance to obstacles
- Battery level
- GPS/location
- Weather conditions

ii. Operating Environment

- Partially Observable
- Dynamic
- Sequential
- Continuous
- Unknown
- Real-world environment

iii. Actions

- Move forward/backward
- Turn
- Collect soil
- Analyze rocks

iv. Performance Evaluation

Performance is measured by:

- Number of samples collected
- Accuracy of analysis
- Distance covered
- Battery efficiency
- Mission completion
- Communication success

v. Suitable Agent Architecture

Utility-Based Learning Agent

Reason:

- Learns from previous observations.
- Makes decisions under uncertainty.
- Chooses actions with maximum utility.
- Suitable for unknown environments like Mars.

Accuracy of Calculation

The PEAS model correctly identifies sensors, actuators, environment, and performance measures.

Interpretation of Results

Mars Rover works as an intelligent autonomous agent capable of sensing, learning, planning, and acting without continuous human control.

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

Sol:

Problem Understanding

Place 8 queens on an 8×8 chessboard so that:

- No two queens share the same row.
- No two queens share the same column.
- No two queens share the same diagonal.

Method Selection

Use Backtracking Search.

Reason:

- Efficiently removes invalid placements.
- Explores only promising solutions.

Solution Process

Initial State

Empty chessboard.

Operators

Place one queen in each column.

Goal Test

Check:

- Same row
- Same column
- Same diagonal

If conflict occurs:

Backtrack.

Continue until all 8 queens are placed.

Example solution:

Column Row

1 1

Column Row

2 5

3 8

4 6

5 3

6 7

7 2

8 4

(No queens attack each other.)

Accuracy of Calculation

Every placement satisfies:

- Different rows
- Different columns
- Different diagonals

Interpretation of Results

Backtracking efficiently searches the state space and finds a valid arrangement without checking every possible configuration.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

Problem Understanding

A customer wants to book a cab.

Options include:

- Mini
- Micro

- Sedan
- Prime
- Shared

The system selects the best cab based on user preference.

Method Selection

Use a Goal-Based Problem Solving Agent.

Reason:

The goal is:

Find the best cab that satisfies user requirements.

Solution Process

Pseudocode

```
START
Input Pickup Location
Input Destination
Display Cab Types
User selects Cab Type
Search available drivers
IF driver available THEN
    Calculate fare
    Display driver details
    Confirm booking
ELSE
    Display "No cab available"
END IF
END
```

Accuracy of Calculation

The algorithm:

- Accepts user input.
- Finds matching cab.
- Books if available.

Interpretation of Results

The Goal-Based Agent searches available cab options and selects one that achieves the user's objective of reaching the destination comfortably.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

Sol:

$S \rightarrow A = 1$

$S \rightarrow G = 12$

$A \rightarrow B = 3$

$A \rightarrow C = 1$

$B \rightarrow D = 3$

$C \rightarrow D = 1$

$C \rightarrow G = 2$

$D \rightarrow G = 3$

Problem Understanding

The logistics company wants to transport packages from warehouse **S** to warehouse **G** at the **minimum travel cost**.

Since every route has a travel cost, the task is to find the **least-cost path**.

Method Selection

Use **Uniform Cost Search (UCS)** because:

- Edge costs are different (weighted graph).
- UCS always expands the node with the **lowest cumulative cost**.
- It guarantees the **optimal (minimum-cost) path**.

Solution Process

Step 1

Start at **S**

Priority Queue:

$$S = 0$$

Step 2

Expand S

Possible paths:

$$A = 1$$

$$G = 12$$

Queue

$$A(1)$$

$$G(12)$$

Step 3

Expand A

New paths

$$B = 1 + 3 = 4$$

$$C = 1 + 1 = 2$$

Queue

$$C(2)$$

$$B(4)$$

$$G(12)$$

Step 4

Expand C

New paths

$$D = 2 + 1 = 3$$

$$G = 2 + 2 = 4$$

Queue

$$D(3)$$

$$B(4)$$

$$G(4)$$

$$G(12)$$

Step 5

Expand D

New path

$$G = 3 + 3 = 6$$

Queue

B(4)

G(4)

G(6)

G(12)

Step 6

Expand B

$$D = 4 + 3 = 7$$

Current best path remains unchanged.

Step 7

Next lowest-cost goal node is:

$$G = 4$$

Search stops.

Shortest Path

$S \rightarrow A \rightarrow C \rightarrow G$

Total Cost

$$S \rightarrow A = 1$$

$$A \rightarrow C = 1$$

$$C \rightarrow G = 2$$

$$\text{Total Cost} = 1 + 1 + 2 = 4$$

Accuracy of Calculation

Path	Cost
$S \rightarrow G$	12

Path	Cost
$S \rightarrow A \rightarrow B \rightarrow D \rightarrow G$	$1+3+3+3 = 10$
$S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$	$1+1+1+3 = 6$
$S \rightarrow A \rightarrow C \rightarrow G$	$1+1+2 = 4$

The **minimum cost is 4**, so UCS correctly finds the optimal route.

Interpretation of Results

Uniform Cost Search successfully identifies the least-cost delivery route:

$S \rightarrow A \rightarrow C \rightarrow G$

with a **minimum total travel cost of 4**. This ensures the logistics company minimizes fuel and travel expenses while guaranteeing the optimal path.